Topic 4 – Natural selection and genetic modification

Students should:		Maths skills
4.1B	Describe the work of Charles Darwin and Alfred Wallace in the development of the theory of evolution by natural selection and explain the impact of these ideas on modern biology	
4.2	Explain Charles Darwin's theory of evolution by natural selection	
4.3	Explain how the emergence of resistant organisms supports Charles Darwin's theory of evolution including antibiotic resistance in bacteria	2c 4a
4.4	Describe the evidence for human evolution, based on fossils, including: a Ardi from 4.4 million years ago b Lucy from 3.2 million years ago c Richard Leakey's discovery of fossils from 1.6 million years ago	1a, 1b, 1c 4a
4.5	Describe the evidence for human evolution based on stone tools, including: a the development of stone tools over time b how these can be dated from their environment	
4.6B	Describe how the anatomy of the pentadactyl limb provides scientists with evidence for evolution	
4.7	Describe how genetic analysis has led to the suggestion of the three domains rather than the five kingdoms classification method	
4.8	Explain selective breeding and its impact on food plants and domesticated animals	
4.9B	Describe the process of tissue culture and its advantages in medical research and plant breeding programmes	
4.10	Describe genetic engineering as a process which involves modifying the genome of an organism to introduce desirable characteristics	
4.11	**Describe the main stages of genetic engineering including the use of:** **a restriction enzymes** **b ligase** **c sticky ends** **d vectors**	
4.12B	Explain the advantages and disadvantages of genetic engineering to produce GM organisms including the modification of crop plants, including the introduction of genes for insect resistance from *Bacillus thuringiensis* into crop plants	

Students should:		Maths skills
4.13B	Explain the advantages and disadvantages of agricultural solutions to the demands of a growing human population, including use of fertilisers and biological control	2c 4a, 4c
4.14	Evaluate the benefits and risks of genetic engineering and selective breeding in modern agriculture and medicine, including practical and ethical implications	2c 4a, 4c

Use of mathematics

- Translate information between numerical and graphical forms (4a).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Plot and draw appropriate graphs, selecting appropriate scales for axes (4a and 4c).
- Extract and interpret information from graphs, charts and tables (2c and 4a).
- Extract and interpret data from graphs, charts, and tables (2c).
- Understand and use direct proportions and simple ratios in genetic crosses (1c).
- Understand and use the concept of probability in predicting the outcome of genetic crosses (2e).